

- Mute state (is it muted?)
- List of channels available
- Current channel setting
- Current volume setting
- Range of volume levels available

And the actions that the TV must provide include:

- Turn the power on and off
- Raise and lower the volume
- Change the channel up and down
- Mute and unmute the sound
- Get information about the current settings
- Go to a specified channel

The code for our `TV` class is shown in [Listing 2-7](#). We include the `__init__()` method for initialization, followed by a method for each of the behaviors.

File: TV.py

```
# TV class

class TV():
    def __init__(self): ❶
        self.isOn = False
        self.isMuted = False
        # Some default list of channels
        self.channelList = [2, 4, 5, 7, 9, 11, 20, 36, 44, 54, 65]
        self.nChannels = len(self.channelList)
        self.channelIndex = 0
        self.VOLUME_MINIMUM = 0    # constant
        self.VOLUME_MAXIMUM = 10  # constant
        self.volume = self.VOLUME_MAXIMUM //    # integer divide

    def power(self): ❷
```